Agency Costs of Free Cash Flow*

Indraneel Chakraborty Vidhi Chhaochharia
Gerardo Pérez-Cavazos Andreya M. Silva

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Abstract

Managers who use Free Cash Flow (FCF) for private benefits impose agency costs on equity-holders. We identify unobservable FCF agency conflicts using firm-level dividend policy changes and executives' personal portfolio decisions, around exogenous tax rate changes, as measures. We find that managers suffering from FCF conflicts take less debt despite tax benefits, plausibly to avoid committing cashflow to interest payments. Due to FCF diversion, such firms also invest less. Capital markets infer FCF conflicts and discount such firms. Finally, we find that firms with large institutional holdings or those with better-aligned executive compensation, suffer less from FCF agency conflicts.

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*Indraneel Chakraborty: School of Business Administration, University of Miami, Coral Gables, FL 33124; Email: i.chakraborty@miami.edu. Vidhi Chhaochharia: School of Business Administration, University of Miami, Coral Gables, FL 33124; Email: vidhi@miami.edu. Gerardo Pérez-Cavazos: Harvard Business School, Boston, MA 02163; Email: gperezca@chicagobooth.edu. Andreya M. Silva: School of Business Administration, University of Miami, Coral Gables, FL 33124; Email: a.silva13@umiami.edu.

Agency conflicts between firm stakeholders are important wedges between optimal and observed firm policy. However, identifying various agency conflicts and measuring their impact on firm policy is difficult. This is because econometricians, just like external stakeholders in a firm, are unable to observe the actions of management. Furthermore, the external stakeholders suffer from information asymmetry, which limits their ability to even ascertain optimal firm policy.

In this paper, we focus on a specific but important agency conflict among specific stakeholders: free cash flow (FCF) conflicts between firm management and equityholders (Jensen, 1986). In firms which have more cash than profitable investment opportunities, managers may prefer diverting such cashflow for private benefits rather than paying it out to equityholders. Jensen (1986) himself suggested the benefits of debt in reducing agency costs of FCF, since interest payments provide a commitment for the FCF and debtholders have incentives to monitor the firm management. It then stands to reason that firms with high FCF agency costs will resist taking on debt, despite the tax-benefits. We estimate the impact of FCF agency conflicts on firm leverage, and ultimately firm value.

The key identification challenge is that agency conflicts are inherently unobservable. To address this challenge, we follow a large body of literature that identifies the impact of unobservable cognitive and non-cognitive skills on social and economic success (See Murnane, Willett, and Levy, 1995; Cawley, Heckman, and Vytlacil, 2001; Heckman, Stixrud, and Urzua, 2006; Cunha and Heckman, 2007, 2008; Cunha, Heckman, and Schennach, 2010, among others). The strategy is to employ measurement equations to identify the impact of unobserved heterogeneity in skill-level on individual outcomes under structural assumptions. In our case, to obtain measurement equations that provide information about unobserved agency conflicts, we exploit changes in U.S. personal dividend and capital gains tax rates in the new millenium.

We utilize five measurement equations due to two tax rate changes in the years 2003 and 2012, and the possible rate increase in the year 2010. The argument we use for the first two measurement equations is that while FCF agency conflicts are unobservable, a key feature is the aversion of managers to return cash to equityholders. Thus, ceteris paribus, firms with high FCF agency costs will disburse relatively less dividends to shareholders – even if dividend tax rates decline – as they did in the year 2003. Hence, the first measure is an indicator variable that denotes if the firm increased dividend payout significantly post 2003 tax rate reduction. Similarly, before the anticipated rise in taxes in 2010, and the actual increase in dividend tax rates starting from 2013, firms on average accelerated dividends to reduce the tax liability of their equity holders (Hanlon and Hoopes, 2014). We document that equityholders value such payouts. Figure 1 shows that firms that announced accelerated dividends experienced a positive cumulative abnormal return (CAR) of approximately 0.66 pp. However, managers with FCF agency conflicts should be less likely to payout accelerated dividends. Hence, the second measure is an indicator which equals one if the firm accelerated dividends in the year 2012.

In addition to dividend policy changes, which may be influenced by firm level characteristics, we utilize purchase/sale decisions of executives in their personal portfolio. These provide us the next three measurement equations. The intuition here is twofold: the personal portfolio decisions are relatively uncorrelated by financing constraints or investment opportunities of the firm. This is even more so given the increasing correlation between individuals stocks and overall market. However, managers who are interested in offloading a large fraction of their equity stake in the firm given the right opportunity, should be less aligned in terms of incentives with the firm. Hence, using equity sales by managers in

2003, 2010, and 2012, we obtain three additional measures of agency conflicts between firm management and equityholders.

We find that the estimated agency costs due to FCF has significant effects on optimal firm policies. Such firms tend to take less leverage despite the tax benefits of debt. Specifically, we find that one unit standard deviation (SD) difference in agency conflicts due to FCF leads to 1.5 unit SD difference in tax rates paid by firms, which is due to 0.26 SD units of under-leverage. Under-leverage is costly to firm equity holders, as the firm pays higher taxes than necessary. We find that firms with one standard deviation higher FCF agency conflicts have 0.62 units lower Tobin's q. This translates to a discount of about a half SD of Tobin's q for unit SD of FCF agency conflicts.

Finally, we show that compensation contracts that provide executives stronger incentive alignment with equityholders reduce FCF agency conflicts. Firms with significantly high institutional ownership do not have statistically or economically significant impact of FCF agency conflicts. This shows that FCF conflicts can be mitigated through various ownership and compensation schemes. We also show that the results are not driven by firm financing constraints, by measure of marginal tax rates or by exact identification strategy.

Our paper contributes to the extensive literature that studies the impact of agency conflicts on firm policy in general, and leverage and firm value in particular. The seminal work of Jensen and Meckling (1976) discusses the relationship between firm management, equity holders and debt holders. Subsequent papers on agency conflicts have analyzed how leverage can help discipline managers by reducing free cash flow (see Myers, 1977; Jensen, 1986, 1989, among others). The literature has also investigated the benefits of leverage on firm value due to tax shielding. Modigliani and Miller (1963), which builds upon the seminal paper Modigliani and Miller (1958), discusses the benefits of tax-shielding in reducing cost of capital of a firm. In another paper, Leland (1994) solves for optimal firm leverage in presence of taxes, default costs, and endogenous default.

Empirically, Parrino and Weisbach (1999) examine the importance of stockholder-bondholder conflicts in capital-structure choices and find sizeable investment distortions. Nikolov and Whited (2014) investigate which agency conflicts affects firm cash policy, and estimate a dynamic model of finance and investment. They find that perquisite consumption has significant impact on cash policy. They also find that size-based compensation matters, but less. We use a structural estimation approach with measurement equations for identification, and investigate the impact of the specific agency conflict of FCF on firm leverage decisions and ultimately on firm value.

Empirical work on optimal leverage seeks to estimate whether firms leave money on the table by under-levering. See the seminal work by Graham (2000) who suggests that firms are not taking the full advantage of tax-shielding benefits and are thus under-leveraged. Recent work by Blouin, Core, and Guay (2010), who use improved estimates of marginal tax rates faced by firms, suggests that firms may not be as under-leveraged. On the other hand, Hennessy and Whited (2005) develop a dynamic trade-off model with endogenous choice of leverage, distributions, and real investment and show that there is no target leverage ratio. Our paper shows that FCF agency conflicts can help explain firm leverage decisions.

Section I discusses the interplay between agency conflicts, tax policy and firm financing policy in more detail. Section II discusses data. Section III reports the results and Section IV discusses robustness tests. Section V concludes.

I Agency conflicts, firm policy and taxes

The central objective of this paper is to measure what fraction of differences in firm policy can be explained by free cash flow problems between managers and equity holders. The agency conflict in question is the private benefits that accrue to firm management due to the availability of free cash flow (Jensen, 1986).

How can we identify agency conflicts between firm management and equity holders due to free cash flow? Exogenous changes in available cash flow due to tax rate changes provide us an opportunity to distinguish managers by level of agency conflicts. If a manager suffers from higher agency conflicts and enjoys free cash flow, and the accompanying private benefits (such as diversion of resources), then we should expect that such a manager will be less inclined to disburse cash to other stake holders. In this paper, we attempt to identify agency conflicts through changes in dividend and capital gains tax policy, and the subsequent optimal dividend policy decisions made by executives.

I.A Identification

The primary hurdle of identifying agency conflicts is that such conflicts are unobservable. Just as the principal (i.e. equity holders) cannot, the econometrician also cannot observe the private decisions of the agent (i.e. firm management). We can only observe the outcome of an agent's decisions, such as firm policy and stock returns which may be partially driven by firm policy. One could try to differentiate between observed firm policies and optimal firm policies. However, firm policy decisions can be made due to a variety of reasons, most of which may be unobservable. This unobservability of private information of managers makes estimation of optimal firm policy difficult. Without identifying optimal firm policy, it is difficult to measure the impact of agency costs on firm policy, and ultimately firm value.

Hence, we utilize a structural estimation approach similar to that in Cunha, Heckman, and Schennach (2010) who estimate multistage production functions for skill formation. Similar to agency conflicts, skills are also unobservable, but measurements are available for children's skill. Although managers and firm equity holders may have a variety of agency conflicts, we focus on the ability of managers to divert cash flow to enjoy private benefits.

We note that after large changes in personal tax rates, managers take decisions regarding their personal portfolio and also make decisions regarding firm payout and financing policy. We expect that when tax rates are expected to increase, managers respond by accelerating dividends (Hanlon and Hoopes, 2014) and by taking more debt because tax-shielding becomes more attractive (Miller, 1977). Similarly, when tax rates decrease, we expect managers to, ceteris paribus, pay out more dividends to firm equity holders (Blouin, Core, and Guay, 2010). We argue that the relative payout ratio of managers in different firms in response to exogenous tax rate changes informs us about incentive alignment between manager and equityholders regarding the use of cash flow.

Regarding their personal portfolios, we expect that managers that have large amounts of capital gains may realize some gains before tax rate increases (Pérez-Cavazos and Silva, 2014). However, we posit that the level of incentive alignment between a manager and the firm declines when managers reduce equity stake in the firm. We argue that the relative fractions of equity sold by managers in different firms conveys information about incentive alignment between managers and equityholders.

In this work, we exploit the large changes in tax rates in 2003 and 2013, and the significant political uncertainty regarding extension of 2003 tax rates beyond 2010. The Jobs and Growth Tax Relief Reconciliation Act of 2003 (JGTRRA) reduced the qualified dividend and long term capital gains tax rates. Conversely, in 2013 the Affordable Care Act imposed a 3.8% personal income tax on investment income for taxpayers with an adjusted gross income (AGI) of $250 thousand for married joint filers (MFJs) and $200 thousand for single filers. More details about tax changes are in Section II.A.

The following insight provides us measures of agency conflicts at firm-manager pair level:

(M1) A manager with higher FCF agency conflicts should be less inclined to pass on higher dividends to equity holders when dividend tax rates decrease, such as in 2003.

(M2) Further, in presence of an anticipated increase in dividend tax rates, as in 2010 and 2012, such a manager should be less inclined to provide a special dividend to equity holders.

Let $Z_{k,t,j}$ be the jth measurement of agency conflicts at time t for manager-firm pair k. We assume that the measurements are additively separable functions of the latent agency conflict factor $\theta_{k,j,t}$, which allows us estimate the following system of equations:

$$\begin{aligned} Z_{k,j,t} &= \mu_j + \alpha j \theta_{k,j,t} + \epsilon_{k,t,j} \\ Y_{k,t,j} &= \kappa + \beta\theta_{k,j,t} + \gamma_{k,j,t} X_{k,j,t} + \eta_{k,t,j}, \end{aligned} \tag{1}$$

where $Y_{k,t,j}$ is the observable variable of interest, and $j \in \{1, \ldots, M\}, t \in \{1, \ldots, T\}$. Error terms $E(\epsilon_{k,t,j}) = 0$, $E(\eta_{k,t,j}) = 0$ are assumed to be uncorrelated across firm-managers, and over time. These assumptions allow us to identify factor loadings α_j. Since we have more than two measurement equations, the factor loadings are identified only upto a normalization, where we normalize $\alpha_1 = 1$. The latent variable θ is scaled to a standard normal distribution. See Cunha, Heckman, and Schennach (2010) for a detailed discussion of the identification strategy.

I.B Testable Hypotheses

Using the responses of firm managers to tax rate changes as a measure of FCF agency conflicts, we estimate the variation in leverage across firms that can be explained by agency conflicts. We then test if the equity markets recognize and discount firms with agency costs due to free cash flow problems. Finally, we estimate the efficacy to various mechanisms that can reduce such conflicts.

The three hypotheses we test are:

(H1) Manifestation of agency conflicts in debt policy: The senior claimant on a firm's cash flow are debt holders, who may monitor the firm and through interest payments reduce free cash flow problem. Thus, if a firm faces agency conflicts, a manager should be less interested in taking debt, even though its tax advantageous. The reason is that the benefits of tax-shielding due to debt accrue to all equity holders, while the private benefits of free cash flow are fully internalized by the firm management.

(H2) Assessment of firm value by markets based on agency conflicts: If managers engage in diversion of cash flow for private benefits due to agency conflicts, we should find that such firms invest less and have lower valuation, i.e., Tobin's q.

(H3) Compensation and monitoring reduce misalignments: Presence of (a) mechanisms that align incentives, such as compensation, or (b) stakeholders with relatively low cost of monitoring, such as large block holders should ameliorate agency conflicts.

The next section describes the data used in this analysis.

II Data

We combine multiple datasets for this project. We obtain data on trading by firm insiders from Thomson Reuters. This database contains corporate insiders' trades, as reported on SEC-required Form 4 filings of insider transactions. We extract all insider acquisitions and dispositions of shares from 1986 through 2012 to reconstruct each insider's portfolio. We merge the insider transaction data with executive data from Execucomp, which covers the top five highest compensated officers of each firm on the S&P 1500. We match each executive's firm with stock return data from CRSP. We also retrieve dividend data from CRSP. Lastly, we match each firm to accounting data from Compustat.

II.A Recent Tax Rate Changes

Given that executives are generally high-earners who pay marginal taxes in the upper tax brackets, we focus on historical changes in the highest personal income statutory tax rates.

Tax rate change of 2001 and 2003

In 2000, the U.S. elected a new U.S. president who had campaigned on reducing tax rates. In June 2001, the Economic Growth and Tax Relief Reconciliation Act of 2001 (EGTRRA) was enacted. EGTRRA gradually lowered income tax rates for the wealthy over a six-year period. In 2001, the highest tax rate decreased from 39.6 pp. to 39.1 pp. In 2002 and 2003, the tax rate declined to 38.6 pp.

In May 2003, the Jobs and Growth Tax Relief Reconciliation Act of 2003 (JGTRRA) was enacted. JGTRRA made three significant changes that impacted high-income taxpayers. First, it accelerated EGTRRA's schedule to reduce taxes. In particular, JGTRRA accelerated a 35 pp. tax rate for high-income taxpayers, which was previously scheduled to take place in 2006. Second, it lowered the long-term capital gains tax rate from 20 pp. to 15 pp. Lastly, the qualified dividend tax rate was reduced to match the capital gains tax rate.

Therefore, in their personal portfolios, we expect that in 2003 executives would accelerate the stock dispositions that they were previously holding to obtain more favorable tax rates on their gains in the future. We use the relative sale of shares to identify incentive alignment between firms and their executives.

Regarding firm policy, the lower dividend tax rates encourage a more generous dividend policy, and again by comparing across firms, after controlling for other determinants, we identify the proclivity of managers to payout cash.

Uncertainty regarding tax rates past 2010

Tax Relief, Unemployment Insurance Reauthorization, and Job Creation Act of 2010 extended the tax rates enacted in EGTRRA and JGTRRA for two more years. However, there was significant political uncertainty about this extension. The compromise agreement between the political parties was achieved as late as December 2010. This uncertainty provided incentives to executives to realize capital gains and dividends before tax rates increased.

Tax rate change in 2013

U.S. enacted significant increases in tax rates for high-income taxpayers starting 2013. First, commencing in 2013, the Patient Protection and Affordable Care Act (PPACA) established a 3.8 pp. personal income tax on investment income, which includes dividends and capital gains (IRC section 469).

Second, the favorable tax provisions set in EGTRRA and JGTRRA expired on December

31st, 2012. To address the impending expiration of tax rates in EGTRRA and JGTRRA, the American Tax Payer Relief Act of 2012 (ATRA) was introduced to the House on July 24th 2012. Given the high degree of political uncertainty, the bill was put on hold during the presidential election and finally signed into law on January 2nd 2013.

The final version of ATRA affected high-income taxpayers by raising the ordinary income rate from 35 pp. to 39.6 pp., the qualified dividend rate from 15 pp. to 20 pp., and the capital gain rate from 15 pp. to 20 pp. Thus, the PPACA and ATRA jointly raised the maximum tax rate on capital gains and dividends from 15 pp. to 23.8 pp.

These tax rate changes provided incentives similar to those in 2010. Moreover, after the tax rates were raised investors also had an incentive to realize losses at a higher tax rate.

II.B Built-in capital gains

To calculate the fraction of capital gains (losses) harvested by executives at different points in time, we first calculate executives' portfolio gains and resulting tax benefits by following Pérez-Cavazos and Silva (2014). Specifically, we quantify each executive's long-term built in capital gains at different points in time by reconstructing each executive's portfolio starting with his first insider transaction as reported by Thomson Reuters Form 4 filings.[1] For each purchase of shares thereafter, we create a transaction record with the number of shares and the corresponding purchase price. This provides us the tax basis. When an executive sells shares, we reduce the number of shares in each batch using the First-In First-Out (FIFO) method.

[1] We eliminate all the transactions that do not meet Thomson Reuter's three highest levels of "clean" records, as well as transactions that do not correspond to an acquisition or disposition of shares. We further remove outliers that surpass the total value of the firm. When calculating the total capital gains, we restrict the minimum number of shares of each executive to zero. Lastly, we truncate the capital gains dataset at the 0.1% and 99.9%.

At each period of interest, we observe the number of shares that remain in the executive's portfolio and the current price per share as reported by CRSP. The tax basis is then subtracted from the current value of the shares to obtain the built-in capital gains of each portfolio. Thus, at each time period the built in capital gains for a particular executive are given by the following equation:

$$\text{Built in capital gains}_t = \sum_{i=1}^{N} (P^t - P_i) \times x_i^t \tag{2}$$

where P^t is the price at time t, P_i is the price at which shares in batch i were acquired, x_i^t is the number of shares in purchase transaction i at time t, and N is the total number of transactions in the executive's portfolio.

II.C Summary Statistics

Table I Panel A reports summary statistics of the dataset. Our identification strategy relies on executives reacting to tax law changes. We find that the response to tax rate changes by firms in terms of dividend policy is significant. Approximately 62% of firms increased their dividend by more than 25% over the three year period due to the tax rate changes in years 2001 and 2003. 14% of firms accelerated their dividends in response to the tax rate increase in year 2013. The response to tax rate changes in the personal portfolio of the executives is also significant. Other firm level variables in our dataset are in line with prior literature. Book leverage is approximately 24%, non-fixed assets are approximately half of firm asserts.

Table I Panel B reports the correlation between the variables of interest. We note that MTRC and TDWL are only weakly correlated in our data. The positive correlation between MTRC and "Large Div?" shows that firms with higher benefits of MTRC increased dividends after tax rate reductions in 2003, which is intuitive. However, firms with higher MTRC did not accelerate dividends significantly in 2012, although the correlation is weak. This maybe because such firms have higher debt to capture tax-shielding benefits, and thus not much additional cash flow is available for early disbursement. Further, Tobin's q is lower for firms which save more through reduction in marginal tax rates. This may again be due to selection bias, because firms that save more through higher MTRC may also be firms that are in more mature industries.

These results suggest the need for a careful analysis that addresses selection bias of firms, and separates the agency conss of FCF from alternative explanations of firm policy. The next section conducts such an investigation.

III Empirical Results

This section starts with a structural estimation of impact of agency conflicts due to FCF on firm leverage (H1). It then investigates the impact of FCF agency conflicts on firm investment and firm value (H2). The section concludes with an analysis of scenarios that reduce FCF agency conflicts (H3).

III.A Measurement equations

The five measurements we use to identify agency conflicts are: (i) firms that increased their dividend significantly in response to the dividend tax rate reductions in the year 2003, (ii) firms that accelerated their dividends in 2012 in response to tax rate increases in 2013, (iii) fraction of capital gains captured by managers out of their portfolio in 2003, (iv) in 2010, and (v) capital gains losses captured in 2013.

The argument behind the first two measures is that in response to dividend and capital gains tax-rate reductions, firm management should increase dividends. In response to tax rate increases, we should expect firms to reduce dividends or accelerate dividends before the tax rate increases. Regarding measurement equations (3)–(5), the argument is that in their personal portfolio, managers who sell more equity when tax-rates are favorable, are less aligned with firm equityholders.

III.B FCF agency conflicts and under-leverage

Managers suffering from FCF agency conflicts would be averse to having a senior claimant to FCF, i.e. debt holders. Thus, such managers will take less debt inspite of the tax benefits of debt. This is because the private benefits of FCF accrue to the manager, while the benefits of tax shielding are not completely internalized by the firm management.

We start by estimating the impact of agency costs of FCF on optimal firm leverage. Blouin, Core, and Guay (2010) develop improved estimates of marginal tax rates (MTR) using a non-parametric procedure. The dataset provides the marginal tax rate pre and post tax-shielding benefits due to interest rate deductions. We calculate the percentage reduction in tax rates $MTRC$ achieved by a firm as the difference between the tax rates of the firm pre and post tax shielding benefits scaled by the tax-rate of the firm pre-tax interest deduction. Thus a firm with 0 MTRC does not avail tax shielding benefits at all, while a firm with 1 MTRC maximizes tax shield benefits, and pays 0 tax rate.

Table II reports the results. Panel A reports the main results and Panel B reports the estimates for the measurement equations. Column (1) conducts the estimates without fixed effects, Column (2) includes year fixed effects and Column (3) includes industry and year fixed effects.

We note that one standard deviation of FCF agency conflicts, based on pooled standard deviation, leads to marginal tax rate difference of 18 pp between firms. In other words, to capture private benefits of FCF, managers who have one standard deviation agency conflicts, are willing to take less debt – leading to firms pay 18% higher tax rate due to under-leverage compared to firms with no agency conflicts.

Column (2) shows that some of these stark estimated effects are attributable to business cycles, as profits of firms vary period by period, and with that relative tax advantages also vary. Thus, including year fixed effects reduces the estimated coefficient of FCF agency conflicts to 12.47%. Finally, some of this estimated effect is also driven by industry level differences. Including industry fixed effects in Column (3) reduces the estimated coefficient to 8.68%. The marginal tax rate paid by firms in the Blouin, Core, and Guay (2010) dataset is 28.28% (SD is 12.48%) before tax shielding and 24.30% (SD is 13.72%) after tax shielding, and the average book leverage of firms is 24% (SD is 41%). Thus, one unit SD difference in agency conflicts due to FCF leads to 1.5 unit SD difference in tax rates paid by firms. In Column (5) we show that one SD change in agency conflicts leads to 10.54% change in book leverage, which is 0.26 SD units of book leverage. All estimated effects of FCF agency costs on marginal tax rate reduction and book leverage are statistically and economically significant.

The identification of FCF agency conflicts is based on the five measurement equations. Estimates of those equations are in Panel B. We normalize the coefficient in the first measurement equation to -1, i.e. firms that raised dividends by more than 25% are firms with relatively least/no agency conflicts. The second measurement equation shows that one standard deviation increase in FCF agency costs reduced the probability of firms accelerating dividends in 2012 by 19 pp. The next two measurement equations show that managers are much more likely to sell shares of their firms – to capture capital gains (lower) at a lower (higher) tax rate – when opportunity arises if they suffer from FCF agency conflicts. When the tax rates declined in 2003, measurement equation (3) suggests that managers with 1% of standard deviation higher agency cost captured 10% more capital gains by selling shares. Measurement equation (4) shows that results are similar in 2010, where there was significant uncertainty about tax rate changes past 2010. Finally, managers also sold more shares in 2013, this time to harvest losses after tax rate increased on capital gains.

These results are consistent with (H1) that firms with high agency conflicts of FCF take lower leverage. Under-leverage is costly to firm equity holders, as the firm pays higher taxes than necessary.

III.C FCF agency conflicts and under-investment

A possible strategy for managers to benefit from FCF is to divert it into projects that are sub-optimal for the firm but provide the managers private benefits (such as empire building). In such a case, hypothesis (H2) suggests that firms should under-invest in PP&E. At the same time, if capital markets can ascertain that such investments are sub-optimal, firm value or Tobin's q should be relatively low compared to the level of investment.

Table III reports the estimated relationship between firm investment and agency conflicts due to FCF. Column (1) reports the OLS results with industry and year fixed effects. Column (2) reports the results for the structural estimation. Again, the specification for investment regression includes industry and year fixed effects, along with Tobin's q, fraction of intangible assets in the firm, size, and operating income before depreciation scaled by assets. We note that firms with one standard deviation (SD) higher agency conflicts invest significantly less. One pp. of a SD higher agency conflicts due to FCF translates to 0.20 pp. lower investment.

Given that the SD of investment in our sample is 9 pp., this means that approximately a half standard deviation increase in agency costs due to FCF leads to a unit SD difference in investment.

Columns (3) and (4) reports an OLS estimation for comparison and the structural estimation of relationship between agency costs and Tobin's q, respectively. We note the statistically and economically significant negative correlation between agency conflicts due to FCF and Tobin's q. Firms with one standard deviation higher FCF agency conflicts have 0.62 units lower Tobin's q. This shows that capital markets discount such firms significantly. Given the mean Tobin's q of 1.9 in our sample, and a SD of 1.1, this translates to a discount of 56% of a standard deviation of Tobin's q.

As before, the identification of FCF agency conflicts is based on the five measurement equations. Estimates of those equations are in Panel B. We again normalize the coefficient in the first measurement equation to -1, i.e. firms that raised dividends by more than 25% are firms with relatively least/no agency conflicts. The remaining measurement equations continue to show that firms where managers accelerate dividends have lower agency conflicts, and firms where managers sell shares due to tax incentives on average have higher agency conflicts.

These results provide evidence in favor of (H2), i.e., firms with more FCF agency costs invest less, and capital markets are able to discern such agency conflicts and discount the firms.

III.D Compensation and blockholders

The previous two sections show that agency costs in FCF can lead to under-leverage and over-investment.

What can be done to ameliorate such conflicts? This section investigates if (i) compensation contracts which align executives' incentives with equity holders, and (ii) large blockholders who have more incentives to monitor the firm reduce FCF agency conflicts.

Table IV reports the relationship between various characteristics of compensation and FCF agency conflicts. All columns include industry and year fixed effects. We obtain data on firm-level executive compensation from the website of L. Naveen (see Core and Guay, 2002; Coles, Daniel, and Naveen, 2006, for more details).[2] Column (1) shows that managers that get paid more relative to the industry have lower agency conflicts. Similarly managers that have higher incentive alignment with the firm due to high delta values, have lower agency conflicts. Column (2) shows that 1% of a standard deviation decrease in FCF agency conflicts is associated with 6 pp. of delta. Thus, the right compensation structure can reduce agency conflicts, but it requires a significant fraction of the firm. This is intuitive because private benefits of FCF diversion are internalized completely by the executives, and hence the compensation to induce the managers not to divert has to be significant as well. Column (3) shows that the same pattern holds for vega term of the compensation structure as well. Thus, managers who have high agency conflicts have low vega in firm compensation. This maybe because low vega reduces the incentive of the manager to take risky projects on behalf of the firm, and managers in place invest in projects with high private benefits.

Next, we focus on firms with above median institutional ownership. Institutional owners with large stakes, and relatively lower costs of monitoring can reduce agency conflicts and its impact on firm policy. Table V reports the impact of agency costs of FCF on firm policy. As Columns (1)–(3) show, the table finds that such agency conflicts have statistically and economically insignificant impact on firms with high institutional ownership. Firms with

[2]The web address is `http://sites.temple.edu/lnaveen/data/`.

high institutional ownership do not seem to take less leverage as measured by MTRC or by TDWL.

As before, the five measurement equations are used to identify agency conflicts in a structural estimation model. These results are consistent with hypothesis (H3), that agency conflicts due to FCF can be ameliorated using better aligned compensation and monitoring by large stakeholders.

IV Robustness

Table VI which uses total dead weight loss in place of MTRC. The results remain similar, providing additional confidence in results of Table II. Table VII use a different set of measurement equations, and find results similar in magnitude to previous results.

An important concern maybe that the identified problem of such firms is not agency conflicts of FCF, but rather constraints regarding financing. There are two reasons why this is not the case. First, the measurement equations are composed of two decisions at firm level, and three at personal portfolio level of the executive. Firm level constraints should not be strongly correlated with executives' personal portfolio decisions. However agency conflicts should be related to personal portfolio decisions, because managers suffering from agency conflicts with the firm should be more inclined to sell equity stakes in the firm given the right opportunity. The identified latent variable is indeed strongly correlated with personal portfolio decisions, and suggests managers sell more equity when the latent variable is strongly present.

Regardless, to further address concerns, in Table VIII, we include a measure for cash constraints in the main specifications. The results remain robust to the inclusion of constraints, suggesting that firm level constraints are not driving the results.

V Conclusion

This paper attempts to identify a specific agency conflict using dividend tax rate changes. We find that FCF agency conflicts have significant impact on important firm policies such as firm leverage and investment. Capital markets are able to discern these frictions, and discount equity valuations of such firms.

We also find that compensation that aligns incentives of firm management with those of equity holders can ameliorate FCF agency conflicts. Moreover, FCF conflicts seem to have little effect on firms with large institutional holdings.

Executive recruitment decisions, compensation contracts, and shareholder ownership structure appear even more crucial to firm prospects in light of our findings.

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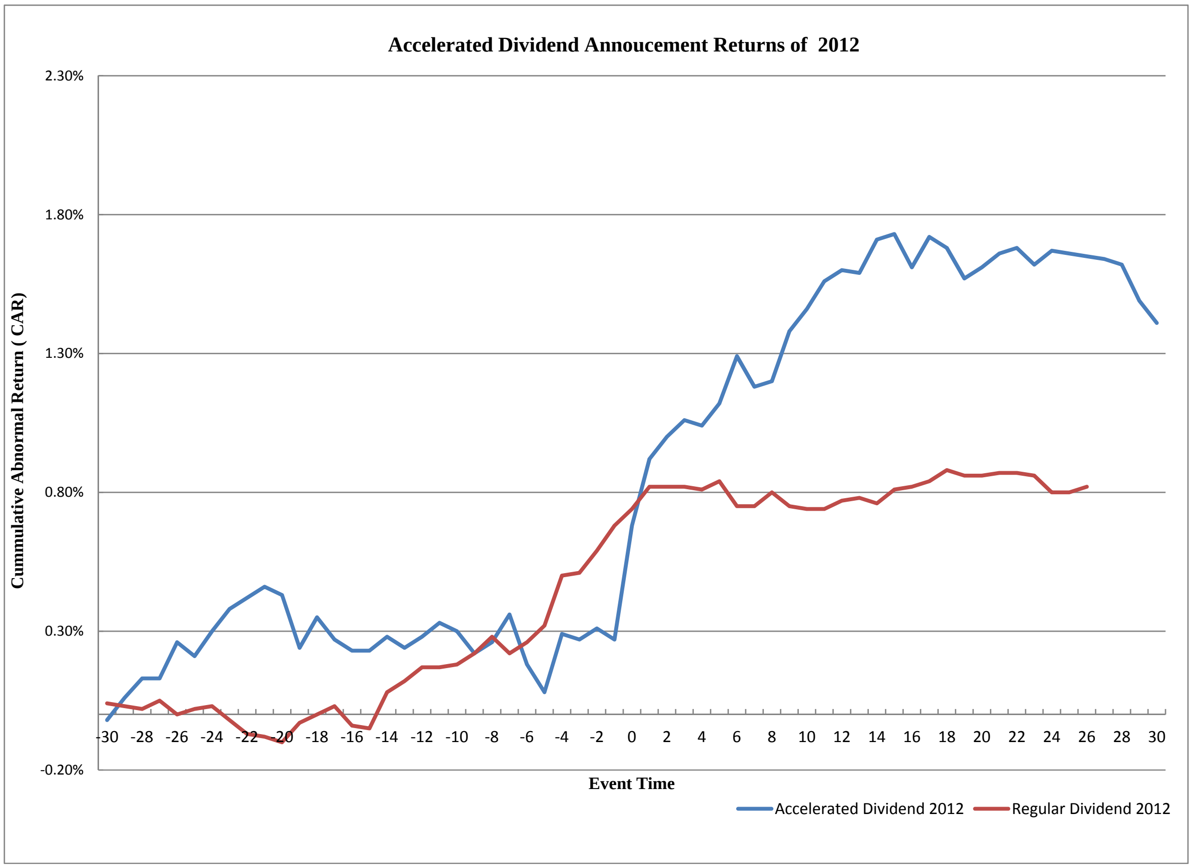
Figure 1: CAR around dividend acceleration announcement



This figure compares cumulative abnormal returns (CAR) of firms that announced accelerated dividends in the year 2012 in response to dividend tax rate increases in 2013, with those firms that did not announce accelerated dividends. The horizontal axis is in days from the announcement. The vertical axis measures equity returns. The number of firms that announced accelerated dividends was 388, whereas the number of firms in the control sample is 3111.

Table I: Descriptive Statistics

This table presents summary statistics. The sample consists of U.S. firms from 1993-2013. All variables are defined in detail in the appendix. All accounting data is obtained from Compustat and are winsorized at the 5% level. Compensation data is obtained from Execucomp. Our main corporate policy variables are the Marginal Tax Rate Change (MTRC) obtained from Blouin, Core, and Guay (2010), Total deadweight Loss due to underleverage as calculated by Van Binsbergen, Graham, and Yang (2010), Investment and Tobins Q. The data for MTRC is from 1993-2010 and total deadweight loss is available for firms from 1993-2007. We use five measurement equations to capture agency costs of Free Cash Flow (FCF). The first measure is Large Dividend change as of 2003 which is an indicator variable that takes a value 1 if the firm paid a dividend larger than 25% as compared to 2002. The second measure is Accelerated Dividend in the year 2012 which is an indicator variable that flags if a dividend to be paid in 2013 is acceleratedd to the last quarter of 2012. The third and fourth measures are Capital Gains fraction in 2003 and 2010. They are the fraction of capital gains realized by executives in 2003 and 2010 in their personal portfolio in response to tax rate changes. The last measure is the Loss 2013 indicator variable, which equals 1 if an executive waits until 2013 to realize a loss in their personal portfolio due to tax rate increases in 2013. The capital gain and loss measures are calculated using the Insider Data from Thomson Reuters (Form 4).

Panel A: Summarized statistics					
	mean	sd	min	max	N
Firm policy variables					
Marginal Tax Rate Change	0.08	0.17	-2.35	1.00	16,641
Total Deadweight Loss (TDWL (UL))	0.01	0.01	0.00	0.30	4,820
Investment	0.13	0.09	0.03	0.33	21,010
Tobin's q	1.90	1.01	0.82	4.46	20,273
Measures					
Large Dividend 2003	0.62	0.48	0.00	1.00	22,641
Accelerated Dividend 2012	0.14	0.35	0.00	1.00	15,368
Cap Gains 2003	30.41	345.60	-0.84	8845.81	16,938
Cap Gains 2010	14.20	67.88	-1.71	953.31	20,222
Cap Loss 2013	4.27	6.65	0.00	43.00	21,511
Additional firm variables					
Book Leverage	0.24	0.41	0.00	52.82	22,461
Free Cash Flow	0.45	0.40	0.00	1.36	21,228
Need for External Fin.	-2.95	3.14	-9.99	1.03	22,204
Intangible Assets	0.52	0.15	0.00	0.99	22,151
Log Assets	7.22	1.83	-6.91	13.59	22,550
Oper. Inc. Before Dep.	0.10	0.08	-0.06	0.25	22,537
Log Compensation	14.01	1.06	0.73	16.44	10,183
Log Delta	12.49	1.92	0.00	17.24	10,183
Log Vega	10.16	3.41	0.00	14.68	10,183

Panel B: Correlation between variables

Variables	TDWL	MTRC	Large Div?	Acc. Div?	Cap Gains 03	Cap Gains 10	Cap Loss 13	Inv.	Tobin's q
TDWL	1.000								
MTRC	0.046	1.000							
Large Div.2003	0.235	0.145	1.000						
Acc. Div.2012	-0.052	-0.080	-0.015	1.000					
Cap Gains 2003	-0.006	0.004	-0.036	0.029	1.000				
Cap Gains 2010	-0.041	-0.014	-0.010	-0.045	-0.010	1.000			
Cap Loss 2013	-0.045	0.042	-0.174	-0.009	-0.003	0.051	1.000		
Investment	0.073	-0.147	0.278	0.026	0.017	-0.018	-0.145	1.000	
Tobin's q	-0.054	-0.154	0.136	0.057	-0.027	0.028	-0.102	0.375	1.000

Table II: FCF Agency Conflicts and Tax Rates

The table reports results for a structural estimation model where the dependent variable is the Marginal Tax Rate Change (MTRC). The sample consists of firms from 1993-2010. Data on the MTRC is collected from Blouin, Core, and Guay (2010). The latent variable being estimated is the Agency Cost due to FCF. Intangible Assets, Log Assets, Operating income before depreciation, Tobin's q are used as control variables in the structural equation. Data on controls are obtained from Compustat. All control variables are winsorized at the 5% level. There are five measurement equations to identify agency costs due to FCF. Their estimates are reported below the estimates for the structural equation. The first measure is Large Dividend change as of 2003 which is an indicator variable that takes a value 1 if the firm paid a dividend larger than 25% as compared to 2002. The second measure is Accelerated Dividend in the year 2012 which is an indicator variable that flags if a dividend to be paid in 2013 is accelerated to the last quarter of 2012. The third and fourth measures are Capital Gains fraction in 2003 and 2010. They are the fraction of capital gains realized by executives in 2003 and 2010 in their personal portfolio in response to tax rate changes. The last measure is the Loss 2013 indicator variable, which equals 1 if an executive waits until 2013 to realize a loss in their personal portfolio due to tax rate increases in 2013. All variables are defined in the Appendix. Standard errors are reported in parentheses below the coefficients. *, **, *** denote significance at the 10%, 5% and 1% level respectively.

Dependent Variable	Marginal tax rate Change (MTRC)								Book Leverage	
	OLS		Structural Estimation							
	(1)		(2)		(3)		(4)		(5)	
Structural Equation:										
Tax Agency Cost			-0.1800	***	-0.1247	***	-0.0868	**	-0.1054	***
			(0.0406)		(0.0369)		(0.0343)		(0.0379)	
Intangible Assets	0.03731	***	0.0875	***	0.0721	***	0.0486	***	0.3183	***
	(0.0095)		(0.0190)		(0.0172)		(0.0158)		(0.0171)	
Log Assets	-0.0094	***	0.0095	***	0.0023		-0.0018		0.01778	***
	(0.0008)		(0.0033)		(0.0029)		(0.0027)		(0.0028)	
Operating Income before Dep.	-0.2551	***	-0.3840	***	-0.3551	***	-0.2846		-0.0902	***
	(0.0161)		(0.0163)		(0.0005)		(0.0140)		(0.0157)	
Tobin's Q	-0.0005		-0.0015	***	-0.0002		0.0006		0.0032	***
	(0.0004)		(0.0005)		(0.0005)		(0.0005)		(0.0005)	
Constant	0.0724	***	-0.0181		-0.0381		-0.0013		-0.3096	***
	(0.0062)		(0.0317)		(0.0286)		(0.0263)		(0.0274)	
Measurement Equation:										
Large Dividend Change 2003										
Tax Agency Cost			-1		-1		-1		-1	
Constant			0.4245	***	0.4254		0.4245		0.4313	***
			(0.0048)		(0.0048)		(0.0048)		(0.0044)	
Accelerated 2012										
Tax Agency Cost			-0.1703	***	-0.1796	***	-0.1869	***	-0.2088	***
			(0.0271)		(0.0279)		(0.0283)		(0.0263)	
Constant			0.1609	***	0.1609	***	0.1609	***	0.1608	***
			(0.0037)		(0.0037)		(0.0037)		(0.0034)	
Capital Gains Fraction 2003										
Tax Agency Cost			11.1720	*	10.7243		10.5539		10.3655	*
			6.4590		(6.5869)		(6.6758)		(6.1903)	
Constant			18.6679	***	18.6679	***	18.6679	***	18.4531	***
			0.9783		(0.9783)		(0.9783)		(0.8838)	
Capital Gains Fraction 2010										
Tax Agency Cost			15.3707	***	15.8975	***	16.1158	***	18.333	***
			(4.4114)		(4.4980)		(4.5588)		(4.3450)	
Constant			15.6466	***	15.6466	***	15.6466	***	15.8716	***
			(0.6738)		(0.6738)		(0.6738)		(0.6228)	
Loss 2013										
Tax Agency Cost			11.7723	***	12.1499	***	12.4172	***	11.9511	***
			0.6895		(0.7175)		(0.7352)		(0.6526)	
Constant			5.0234	***	5.0234	***	5.0234	***	5.0134	***
			0.0736		(0.7356)		(0.0735)		(0.0672)	
Industry Fixed Effects	Y		N		N		Y		Y	
Year Fixed Effects	Y		N		Y		Y		Y	
N	16331		9510		9510		9510		114433	

Table III: FCF agency costs, firm investment and firm valuation

The table reports results for a structural estimation model where the dependent variables are firm investment and Tobin's q. The latent variable is the Agency Cost due to FCF. Intangible Assets, Log Assets, Operating income before depreciation, Tobin's q are used as control variables in the structural equation. Data on controls are obtained from Compustat. All control variables are winsorized at the 5% level. There are five measurement equations to identify agency costs due to FCF. Their estimates are reported below the estimates for the structural equation. The first measure is Large Dividend change as of 2003 which is an indicator variable that takes a value 1 if the firm paid a dividend larger than 25% as compared to 2002. The second measure is Accelerated Dividend in the year 2012 which is an indicator variable that flags if a dividend to be paid in 2013 is accelerated to the last quarter of 2012. The third and fourth measures are Capital Gains fraction in 2003 and 2010. They are the fraction of capital gains realized by executives in 2003 and 2010 in their personal portfolio in response to tax rate changes. The last measure is the Loss 2013 indicator variable, which equals 1 if an executive waits until 2013 to realize a loss in their personal portfolio due to tax rate increases in 2013. All variables are defined in the Appendix. Standard errors are reported in parentheses below the coefficients. *, **, *** denote significance at the 10%, 5% and 1% level, respectively.

Dependent Variable	Investment				Tobin's q			
	OLS (1)		Struct. Est. (2)		OLS (3)		Struct. Est. (4)	
Tax Agency Cost			-0.2139	***			-0.6165	**
			(0.0359)				(0.2329)	
Intangible Assets	-0.0569	***	0.0152		-1.4209	***	-0.8265	***
	(0.0046)		(0.0139)		(0.0464)		(0.0899)	
Log Assets	-0.0038	***	0.0172	***	0.0050		0.0763	***
	(0.0003)		(0.0027)		(0.0036)		(0.0162)	
Operating Income before Dep.	0.0354	***	0.0916	***	1.5328	***	3.4802	***
	(0.0062)		(0.0107)		(0.1563)		(0.0781)	
Tobin's Q	0.0046	***	0.0002		0.1008	***	0.0768	***
	(0.0002)		(0.0005)		(0.0037)		(0.0033)	
Constant	0.0414	***	-0.1649	***	0.3144	***	-0.7158	
	(0.0031)		(0.0258)		(0.0351)		(0.1568)	
Measurement Equation:								
Large Dividend Change 2003								
Tax Agency Cost			-1				-1	
Constant			0.4334	***			0.4255	
			(0.0046)				(0.0046)	
Accelerated 2012								
Tax Agency Cost			-0.1784				0.2626	***
			(0.0238)				(0.0295)	
Constant			0.1608	***			0.1551	***
			(0.0035)				(0.0035)	
Capital Gains Fraction 2003								
Tax Agency Cost			7.4919				5.0427	
			(5.6914)				(6.0653)	
Constant			18.4594	***			16.7863	***
			(0.9181)				(0.8232)	
Capital Gains Fraction 2010								
Tax Agency Cost			21.8812	***			21.2483	***
			(4.0599)				(4.8157)	
Constant			16.0323	***			15.9781	***
			(0.6494)				(0.6509)	
Loss 2013								
Tax Agency Cost			10.6238	***			13.2319	***
			(0.5634)				(0.7190)	
Constant			5.0412	***			4.9714	***
			(0.0701)				(0.0684)	
Industry Fixed Effects	Y		Y		Y		Y	
Year Fixed Effects	Y		Y		Y		Y	
N	19671		10683		18950		10487	

Table IV: FCF agency costs and executive compensation

The table reports results for a structural estimation model where the dependent variables are various compensation contract variables: (i) logarithm of total compensation of the firm's Chief Executive Officer (CEO), (ii) logarithm of CEO delta (iii) logarithm of CEO vega. The sample consists of firms from 1993-2010. The latent variable is the Agency Cost due to FCF. Intangible Assets, Log Assets, Operating income before depreciation, Tobin's q are used as control variables in the structural equation. Data on controls are obtained from Compustat. All control variables are winsorized at the 5% level. There are five measurement equations to identify agency costs due to FCF. Their estimates are reported below the estimates for the structural equation. The first measure is Large Dividend change as of 2003 which is an indicator variable that takes a value 1 if the firm paid a dividend larger than 25% as compared to 2002. The second measure is Accelerated Dividend in the year 2012 which is an indicator variable that flags if a dividend to be paid in 2013 is accelerated to the last quarter of 2012. The third and fourth measures are Capital Gains fraction in 2003 and 2010. They are the fraction of capital gains realized by executives in 2003 and 2010 in their personal portfolio in response to tax rate changes. The last measure is the Loss 2013 indicator variable, which equals 1 if an executive waits until 2013 to realize a loss in their personal portfolio due to tax rate increases in 2013. All variables are defined in the Appendix. Standard errors are reported in parentheses below the coefficients. *, **, *** denote significance at the 10%, 5% and 1% level, respectively.

Dependent Variable	Compensation (1)		Delta (2)		Vega (3)	
Tax Agency Cost	-0.5628	*	-6.7762	***	-3.2263	**
	(0.3391)		(1.8683)		(1.3070)	
Intangible Assets	0.5093	***	1.9296	**	1.6842	***
	(0.1493)		(0.7820)		(0.5834)	
Log Assets	0.2858	***	0.9064	***	0.938	***
	(0.0262)		(0.1446)		(0.1019)	
Operating Income before Dep.	1.5365	***	4.0635	***	2.8079	***
	(0.1267)		(0.4369)		(0.4695)	
Tobin's Q	0.0136	***	0.0254	**	0.0461	***
	(0.0036)		(0.0125)		(0.0133)	
Constant	-2.5782	***	-8.4534	***	-8.3839	***
	(0.2676)		(1.5025)		(1.0546)	
Measurement Equation:						
Large Dividend Change 2003						
Tax Agency Cost	-1		-1		-1	
Constant	0.3897	***	0.3897	***	0.3897	***
	(0.0058)		(0.0058)		(0.0058)	
Accelerated 2012						
Tax Agency Cost	-0.1521	***	-0.1984	***	-0.1379	***
	(0.0350)		(0.0315)		(0.0345)	
Constant	0.1494	***	0.1494	***	0.1494	***
	(0.0044)		(0.0044)		(0.0044)	
Capital Gains Fraction 2003						
Tax Agency Cost	31.1514	***	17.1304	**	36.2879	***
	(9.1264)		(8.0651)		(9.1655)	
Constant	18.5879	***	18.5879	***	18.5879	***
	(1.2040)		(1.2048)		(1.2036)	
Capital Gains Fraction 2010						
Tax Agency Cost	25.6317	***	22.5691	***	24.0995	***
	(5.8561)		(5.2029)		(5.7601)	
Constant	15.9508	***	15.9508	***	15.9508	***
	(0.7878)		(0.7879)		(0.7879)	
Loss 2013						
Tax Agency Cost	13.5444	***	12.2968	***	13.0669	***
	(0.9513)		(0.8139)		(0.9274)	
Constant	5.6074	***	5.6074	***	5.6074	***
	(0.0939)		(0.0939)		(0.0939)	
Industry Fixed Effects	Y		Y		Y	
Year Fixed Effects	Y		Y		Y	
N	6559		6559		6559	

Table V: FCF agency conflicts in firms with high institutional holdings

The table reports results for a structural estimation model where the dependent variables are the Marginal Tax Rate Change (MTRC), Total Deadweight Loss (TDWL) and Total Deadweight Loss from under-leverage (TDWL UL). The table only includes firms with above median institutional holdings. The sample for MTRC consists of firms from 1993-2010. Data on the MTRC is collected from Blouin, Core, and Guay (2010). Data on total dead weight loss variables are obtained from Van Binsbergen, Graham, and Yang (2010) and is from 1993-2007. The latent variable is the Agency Cost due to FCF. Intangible Assets, Log Assets, Operating income before depreciation, Tobin's q are used as control variables in the structural equation. Data on controls are obtained from Compustat. All control variables are winsorized at the 5% level. There are five measurement equations to identify agency costs due to FCF. Their estimates are reported below the estimates for the structural equation. The first measure is Large Dividend change as of 2003 which is an indicator variable that takes a value 1 if the firm paid a dividend larger than 25% as compared to 2002. The second measure is Accelerated Dividend in the year 2012 which is an indicator variable that flags if a dividend to be paid in 2013 is accelerated to the last quarter of 2012. The third and fourth measures are Capital Gains fraction in 2003 and 2010. They are the fraction of capital gains realized by executives in 2003 and 2010 in their personal portfolio in response to tax rate changes. The last measure is the Loss 2013 indicator variable, which equals 1 if an executive waits until 2013 to realize a loss in their personal portfolio due to tax rate increases in 2013. All variables are defined in the Appendix. Standard errors are reported in parentheses below the coefficients. *, **, *** denote significance at the 10%, 5% and 1% level, respectively.

Dependent Variable	MTRC (1)		TDWL (2)		TDWL (UL) (3)	
Structural Equation:						
Tax Agency Cost	-0.0234		0.0043		-0.0019	
	(0.0162)		(0.0072)		(0.0022)	
Intangible Assets	0.0124		-0.0026		0.0081	
	(0.0151)		(0.0049)		(0.0019)	
Log Assets	0.0000		-0.0018	**	-0.0008	***
	(0.0021)		(0.0008)		(0.0002)	
Operating Income before Dep.	-0.1221	***	-0.0165	**	-0.0163	***
	(0.0205)		(0.0080)		(0.0035)	
Tobin's Q	0.0012	*	0.0001		0.0000	
	(0.0006)		(0.0002)		(0.0001)	
Constant	-0.003		0.016	**	0.0027	
	(0.0181)		(0.0066)		(0.0021)	
Measurement Equation:						
Large Dividend Change 2003						
Tax Agency Cost	-1		-1		-1	
Constant	0.4815	***	0.4452	***	0.5032	***
	(0.0088)		(0.0109)		(0.0147)	
Accelerated 2012						
Tax Agency Cost	-0.1366	***	-0.2016	***	-0.0812	
	(0.0435)		(0.0655)		(0.0886)	
Constant	0.1438	***	0.1501	***	0.1453	***
	(0.0064)		(0.0081)		(0.0106)	
Capital Gains Fraction 2003						
Tax Agency Cost	45.2281	***	43.4369	***	68.7654	***
	(5.7377)		(8.7439)		(16.0333)	
Constant	11.9355	***	11.5777	***	12.8474	***
	(0.8030)		(1.0721)		(1.7626)	
Capital Gains Fraction 2010						
Tax Agency Cost	31.2074	***	57.9517	***	73.4958	***
	(6.9840)		(10.3100)		(19.1009)	
Constant	14.7923	***	14.6669	***	15.0975	***
	(0.9746)		(1.0596)		(1.3923)	
Loss 2013						
Tax Agency Cost	13.7598	***	18.9083	***	14.6829	***
	(1.3621)		(2.2485)		(2.8017)	
Constant	4.8118	***	5.0103	***	4.1807	***
	(0.1323)		(0.1693)		(0.1923)	
Industry Fixed Effects	Y		Y		Y	
Year Fixed Effects	Y		Y		Y	
N	3024		1945		1101	

Table VI: FCF Agency Conflicts and Total Deadweight Loss due to Underleverage

This table reports results for a structural estimation model where the dependent variable is the Total Deadweight Loss due to under-leverage. The sample consists of firms from 1993-2007. Data on total deadweight loss is obtained from Van Binsbergen, Graham, and Yang (2010). The latent variable is the Agency Cost due to FCF. Intangible Assets, Log Assets, Operating income before depreciation, Tobin's q are used as control variables in the structural equation. Data on controls are obtained from Compustat. All control variables are winsorized at the 5% level. There are five measurement equations to identify agency costs due to FCF. Their estimates are reported below the estimates for the structural equation. The first measure is Large Dividend change as of 2003 which is an indicator variable that takes a value 1 if the firm paid a dividend larger than 25% as compared to 2002. The second measure is Accelerated Dividend in the year 2012 which is an indicator variable that flags if a dividend to be paid in 2013 is accelerated to the last quarter of 2012. The third and fourth measures are Capital Gains fraction in 2003 and 2010. They are the fraction of capital gains realized by executives in 2003 and 2010 in their personal portfolio in response to tax rate changes. The last measure is the Loss 2013 indicator variable, which equals 1 if an executive waits until 2013 to realize a loss in their personal portfolio due to tax rate increases in 2013. All variables are defined in the Appendix. Standard errors are reported in parentheses below the coefficients. *, **, *** denote significance at the 10%, 5% and 1% level, respectively.

Dependent Variable	Total Dead Weight Loss (Under Leverage)			
	OLS	Structural Estimation		
	(1)	(2)	(3)	(4)
Tax Agency Cost		0.0438 ***	-0.0396 ***	-0.0337 ***
		(0.0143)	(0.0129)	(0.0117)
Intangible Assets	0.0078 ***	0.0334 ***	0.0306 ***	0.02556 ***
	(0.0017)	(0.0099)	(0.0091)	(0.0080)
Log Assets	-0.0013 ***	0.0016 *	0.0010	0.0009
	(0.0001)	(0.0009)	(0.0008)	(0.0007)
Operating Income before Dep.	-0.0493 ***	-0.0645 ***	-0.0629	-0.0509 ***
	(0.0088)	(0.0056)	(0.0052)	(0.0048)
Tobin's Q	0.0000	-0.0003 **	-0.0001	-0.0002 *
	(0.0001)	(0.0001)	(0.0001)	(0.0001)
Constant	0.0105 ***	-0.0132	-0.0181 *	-0.0157 *
	(0.0017)	(0.0112)	(0.0101)	(0.0090)
Measurement Equation:				
Large Dividend Change 2003				
Tax Agency Cost		-1	-1	-1
Constant		0.4359 ***	0.4358 ***	0.4359 ***
		(0.0082)	(0.0082)	(0.0082)
Accelerated 2012				
Tax Agency Cost		-0.0469	-0.0458	-0.0525
		(0.0382)	(0.0384)	(0.0407)
Constant		0.1599 ***	0.1599 ***	0.1599 ***
		(0.0064)	(0.0063)	(0.0064)
Capital Gains Fraction 2003				
Tax Agency Cost		-2.7293	-2.1310	-4.657
		(13.1461)	(13.2069)	(14.0110)
Constant		25.2804 ***	25.2804 ***	25.2804 ***
		(2.1833)	(2.1833)	(2.1833)
Capital Gains Fraction 2010				
Tax Agency Cost		27.3525 ***	26.6214 ***	27.4026 ***
		(6.4892)	(6.5052)	(6.8803)
Constant		16.5851 ***	16.5851 ***	16.5851 ***
		(1.0999)	(1.1000)	(1.0999)
Loss 2013				
Tax Agency Cost		8.5556 ***	8.3991 ***	9.0809 ***
		(0.8748)	(0.8865)	(0.9670)
Constant		4.4426 ***	4.4426 ***	4.4426 ***
		(0.1149)	(0.1149)	(0.1146)
Industry Fixed Effects	Y	N	N	Y
Year Fixed Effects	Y	N	Y	Y
N	4770	3283	3283	3283

Table VII: Robustness: Alternative Measurement Equations

The table conducts a robustness test on the main results (Table II) using a different set of measures. It reports results for a structural estimation model where the dependent variables are Marginal Tax Rate Change (MTRC), Total Deadweight Loss (TDWL), Total Deadweight Loss from under-leverage (TDWL UL) and Book Leverage. The sample for MTRC consists of firms from 1993-2010. Data on the MTRC is collected from Blouin, Core, and Guay (2010). Data on total dead weight loss variables are obtained from Van Binsbergen, Graham, and Yang (2010) and is from 1993-2007. The latent variable is the Agency Cost due to FCF. Intangible Assets, Log Assets, Operating income before depreciation, Tobin's q are used as control variables in the structural equation. Data on controls are obtained from Compustat. All control variables are winsorized at the 5% level. There are five measurement equations to identify agency costs due to FCF. Their estimates are reported below the estimates for the structural equation. The first measure is Large Dividend change as of 2003 which is an indicator variable that takes a value 1 if the firm paid a dividend larger than 25% as compared to 2002. The second measure is Accelerated Dividend in the year 2012 which is an indicator variable that flags if a dividend to be paid in 2013 is accelerated to the last quarter of 2012. The third and fourth measures are Capital Gains fraction in 2003 and 2010. They are the fraction of capital gains realized by executives in 2003 and 2010 in their personal portfolio in response to tax rate changes. The last measure is the Loss 2013 indicator variable, which equals 1 if an executive waits until 2013 to realize a loss in their personal portfolio due to tax rate increases in 2013. All variables are defined in the Appendix. Standard errors are reported in parentheses below the coefficients. *, **, *** denote significance at the 10%, 5% and 1% level, respectively.

Dependent Variable	MTRC (1)		TDWL (2)		TDWL (UL) (3)		Book Lev (4)	
Structural Equation:								
Tax Agency Cost	-0.0573	**	-0.0135	*	-0.0227	***	-0.07331	***
	(0.0246)		(0.0071)		(0.0049)		(0.0237)	
Intangible Assets	0.0433	***	0.0092	**	0.0194	***	0.3138	***
	(0.0148)		(0.0045)		(0.0040)		(0.0153)	
Log Assets	-0.0042	**	-0.0009		0.0002		0.01533	***
	(0.0019)		(0.0006)		(0.0003)		(0.0018)	
Operating Income before Dep.	-0.2838	***	-0.0457	***	-0.0491	***	-0.0843	***
	(0.0138)		(0.0052)		(0.0036)		(0.0154)	
Tobin's Q	0.0005		0.0002	*	-0.0002	**	0.0031	***
	(0.0005)		(0.0001)		(0.0001)		(0.0005)	
Constant	0.0196		0.0039		-0.0071	*	-0.289	***
	(0.0198)		(0.0057)		(0.0039)		(0.0187)	
Measurement Equation:								
Large Dividend Change 2003								
Tax Agency Cost	-1		-1		-1		-1	
Constant	0.4245	***	0.4156	***	0.4359	***	0.4313	***
	(0.0048)		(0.0064)		(0.0082)		(0.0044)	
Accelerated 2012								
Tax Agency Cost	-0.1939	***	-0.1601	***	-0.0604		-0.1982	***
	(0.0279)		(0.0364)		(0.0393)		(0.0251)	
Constant	0.1609	***	0.1665	***	0.1599	***	0.1608	***
	(0.0038)		(0.0050)		(0.0064)		(0.0034)	
Capital Gains Fraction 2003								
Tax Agency Cost	9.5383		4.9747		7.0425		10.7039	*
	(6.4811)		(9.7673)		(13.3787)		(5.8638)	
Constant	18.6679	***	20.9245	***	25.2804	***	18.4531	***
	(0.9784)		(1.4173)		(2.1833)		(0.8838)	
Capital Gains Fraction 2010								
Tax Agency Cost	18.5026	***	30.0609	***	30.3579	***	20.132	***
	(4.4369)		(6.1251)		(6.6426)		(4.1201)	
Constant	15.6466	***	16.4903	***	16.5851	***	15.8716	***
	(0.6736)		(0.8783)		(1.0995)		(0.6227)	
Loss 2010								
Tax Agency Cost	12.3219	***	13.1909	***	9.8113	***	11.7619	***
	(0.7519)		(1.0008)		(0.9714)		(0.6342)	
Constant	5.9762	***	5.9059	***	5.5885	***	5.9203	***
	(0.0774)		(0.1006)		(0.1225)		(0.0703)	
Industry Fixed Effects	Y		Y		Y		Y	
Year Fixed Effects	Y		Y		Y		Y	
N	9510		5476		3283		114433	

Table VIII: Robustness: Agency costs and financial constraints

The table conducts a robustness test on the main results (Table II) by including financial constraints as an alternative explanatory variable. It reports results for a structural estimation model where the dependent variables are Marginal Tax Rate Change (MTRC), Total Deadweight Loss from under-leverage (TDWL UL), Firm Investment and Tobin's q. The sample for MTRC consists of firms from 1993-2010. Data on the MTRC is collected from Blouin, Core, and Guay (2010). Data on total dead weight loss variables are obtained from Van Binsbergen, Graham, and Yang (2010) and is from 1993-2007. The latent variable is the Agency Cost due to FCF. Intangible Assets, Log Assets, Operating income before depreciation, Tobin's q are used as control variables in the structural equation. Data on controls are obtained from Compustat. All control variables are winsorized at the 5% level. There are five measurement equations to identify agency costs due to FCF. Their estimates are reported below the estimates for the structural equation. The first measure is Large Dividend change as of 2003 which is an indicator variable that takes a value 1 if the firm paid a dividend larger than 25% as compared to 2002. The second measure is Accelerated Dividend in the year 2012 which is an indicator variable that flags if a dividend to be paid in 2013 is accelerated to the last quarter of 2012. The third and fourth measures are Capital Gains fraction in 2003 and 2010. They are the fraction of capital gains realized by executives in 2003 and 2010 in their personal portfolio in response to tax rate changes. The last measure is the Loss 2013 indicator variable, which equals 1 if an executive waits until 2013 to realize a loss in their personal portfolio due to tax rate increases in 2013. All variables are defined in the Appendix. Standard errors are reported in parentheses below the coefficients. *, **, *** denote significance at the 10%, 5% and 1% level, respectively.

Dependent Variable	MTRC		TDWL UL		Investment		Tobin's q	
	(1)		(2)		(3)		(4)	
Structural Equation:								
Tax Agency Cost	-0.0999	***	-0.0376	***	-0.2699	***	-0.5301	***
	(0.0356)		(0.0140)		(0.0456)		(0.2296)	
Need for external financing	0.0023	***	0.0006	***	0.0106	***	-0.0088	***
	(0.0006)		(0.0002)		(0.0006)		(0.0034)	
Intangible Assets	0.0637	***	0.0294	***	0.0923	***	-0.8649	***
	(0.0177)		(0.0097)		(0.0192)		(0.0971)	
Log Assets	-0.0018		0.0012		0.0186	***	0.0685	***
	(0.0028)		(0.0008)		(0.0032)		(0.0154)	
Operating Income before Dep.	-0.2570	***	-0.0449	***	0.2178	***	3.4402	***
	(0.0162)		(0.0050)		(0.0142)		(0.0798)	
Lagged Tobin's Q	0.0006		-0.0002	*	0.00003		0.0793	***
	(0.0004)		(0.0001)		0.0006		(0.0034)	
Constant	-0.0059		-0.0180	*	-0.1986	***	-0.6661	***
	(0.0272)		(0.0104)		(0.0317)		(0.1514)	
Measurement Equation:								
Large Dividend Change 2003								
Tax Agency Cost	-1		-1		-1		-1	
Constant	0.4269	***	0.4359	***	0.4328	***	0.4277	***
	(0.0049)		(0.0083)		(0.0045)		(0.0044)	
Accelerated 2012								
Tax Agency Cost	-0.2033	***	-0.0877	***	-0.2055	***	-0.2792	***
	(0.0283)		(0.0423)		(0.0225)		(0.0290)	
Constant	0.1619	***	0.1606	***	0.1608	***	0.1557	***
	(0.0038)		(0.0064)		(0.0035)		(0.0035)	
Capital Gains Fraction 2003								
Tax Agency Cost	6.7546		12.5569		-10.169		-5.214	
	(6.6660)		(14.4192)		(7.2766)		(10.4686)	
Constant	18.6315	***	25.4281	***	25.2804	***	18.6418	***
	(0.9906)		(2.1969)		(2.1833)		(1.4208)	
Capital Gains Fraction 2010								
Tax Agency Cost	18.3153	***	29.7971	***	26.8764	***	24.4337	***
	(4.5710)		(7.0982)		(3.9231)		(4.8055)	
Constant	15.7848	***	16.6632	***	16.0609	***	16.1716	***
	(0.6855)		(1.1065)		(0.6358)		(0.6491)	
Loss 2013								
Tax Agency Cost	12.4219	***	9.8034	***	10.5556	***	13.2111	***
	(0.7135)		(1.0134)		(0.5295)		(0.7076)	
Constant	5.0588	***	4.4561	***	5.0401	***	5.002	***
	(0.0745)		(0.1148)		(0.0686)		(0.0678)	
Industry Fixed Effects	Y		Y		Y		Y	
Year Fixed Effects	Y		Y		Y		Y	
N	9340		3262		11186		10786	

Appendix

A Corporate Policy Variables

- **Marginal Tax Rate (MTRC):** We calculate the percentage reduction in tax rates MTRC achieved by a firm as the difference between the tax rates of the firm pre- and post-tax shielding benefits scaled by the tax-rate of the firm pre-tax interest deduction. The data is obtained from Blouin, Core, and Guay (2010) who develop improved estimates of marginal tax rates (MTR) using a non-parametric procedure. The dataset provides the marginal tax rate pre and post tax shielding benefits due to interest rate deductions. The data is annual and from 1993-2010.

- **Total Deadweight Loss from Under-Leverage (TDWL_UL):** This measure is defined for firms where the equilibrium factor is >1. The equilibrium factor is defined as the intersection of the marginal benefit of debt and marginal cost of debt curves. If eqfac=1, the firm's equilibrium interest expense over book value (IOB) is 100% of its actual (firm is at equilibrium). If eqfac=1.2, the firm should have 120% of its actual IOB (firm is underlevered). If eqfac=0.8, the firm should have 80% of its actual IOB (firm is overlevered). The equilibrium factor is bound at 10 (1000%). The data is obtained from Van Binsbergen, Graham, and Yang (2010). The data are annual and are available from 1993-2007.

- **Total Deadweight Loss (TDWL):** This variable equals tdwl_ol for overlevered firms and tdwl_ul for underlevered firms. Tdwl_ol is the total deadweight loss from overlevering (for firms with eqfac¡1). The data is obtained from Van Binsbergen, Graham, and Yang (2010). The data are annual and are available from 1993-2007.

- **Investment:** (Capital Expediture on PPE (capxv) -Sale of Property (sppe))/ Lagged Gross PPE (ppegt). Data is annual and obtained from Compustat. The sample period is from 1993-2013. Investment is winsorized at the 5 % level.

- **Tobin's q:** (Assets(at) + Common Shares Outstanding (csho) *Close Price at end of fiscal year(prcc_f) - Common Equity(ceq) - Deferred Taxes (txdb)) /Assets (at). Data is annual and obtained from Compustat. The sample period is from 1993-2013. The variable is winsorized at the 5% level.

B Measures

- **Large Dividend 2003:** An indicator variable that takes a value 1 if a firm paid a dividend larger than 25% as compared to 2002. The data is collected from Compustat.

- **Accelerated Dividend 2013:** An indicator variable that takes a value 1 if the firm pays an accelerated dividend is the last quarter of 2012. A company is defined as an accelerator if the company paid a dividend in the first quarter of 2011 and 2012; the company decreased or did not pay a dividend in the first quarter of 2013 and the company increased a payment in the last quarter of 2012. The data is collected from Compustat.

- **Capital Gains 2003:** Fraction of capital gains on June 1st 2003 divided by capital gains on March 1, 2003. We calculate each executive's long-term built in capital gains at different points in time by reconstructing each executive's portfolio starting with his first insider transaction as reported by Thomson Reuters Form 4 filings.When calculating the total capital gains, we restrict the minimum number of shares of each executive to zero. For each purchase of shares thereafter, we create a transaction record with the number of shares and the corresponding purchase price. This provides us the tax basis. When an executive sells shares, we reduce the number of shares in each batch using the First-In First-Out (FIFO) method. The measure is calculated for all executives for which data is available in Thomson Reuters Insider Trading Database that we can merge with Execucomp. We take the maximum value for all executives in a firm as our firm level measure capital gains.

- **Capital Gains 2010:** Capital Gains as of 2010 is calculated as the capital gains on January 1st 2011 divided by capital gains on April 1, 2010. We calculate each executive's long-term built in capital gains at different points in time by reconstructing each executive's portfolio starting with his first insider transaction as reported by Thomson Reuters Form 4 filings.When calculating the total capital gains, we restrict the minimum number of shares of each executive to zero. For each purchase of shares thereafter, we create a transaction record with the number of shares and the corresponding purchase price. This provides us the tax basis. When an executive sells shares, we reduce the number of shares in each batch using the First-In First-Out (FIFO) method. The measure is calculated for all executives for which data is available in Thomson Reuters Insider Trading Database that we can merge with Execucomp. We take the maximum value for all executives in a firm as our firm level measure capital gains.

- **Loss 2013:** We first calculate capital gains as of 2013 as the fraction of gains made on January 1st 2013 divided by capital gains on December 1, 2012. Tax minded executives with a gain would sell their shares (in anticipation of an increase in taxes) so these executives would have capital gains less than 1. If the executive takes a loss they should hold their shares before the tax rate increases. Therefore the capital gains fraction as of 2013 would be greater than 1. Loss 2013 flags executives that have a capital gains fraction larger than 1. The measure is calculated for all executives for which data is available in Thomson Reuters Insider Trading Database that we can merge with Execucomp. We take the sum of this variable for all executives in a firm to create a firm level measure of losses.

C Control and additional firm variables

- **Book Leverage:** Total debt (td)/Assets (at). Data is annual and obtained from Compustat. The sample period is from 1993-2013. Book leverage is winsorized at the 5% level.

- **Free Cash Flow:** Operating Income Before Depreciation (oibdp)/Lagged gross PPE (ppegt). Data is annual and obtained from Compustat. The sample period is from 1993-2013. Book leverage is winsorized at the 5% level.

- **Need for External Financing:** (Capital expenditure (Capxv) -Funds from operations (fopt)) /Capital expenditure. Data is annual and obtained from Compustat. The sample period is from 1993-2013. External financing is winsorized at the 5% level.

- **Intangible Assets:** Tangibility is defined as (0.715*Total Receivables (rect) + 0.547*Inventories(invt) + 0.535*Net PPE(ppent) + Cash and Short term Investments (che))/Assets (at). Data is annual and obtained from Compustat. The sample period is from 1993-2013. Intangible Assets is winsorized at the 5% level.

- **Log Assets:** Log (Assets (at)). Data is annual and obtained from Compustat. The sample period is from 1993-2013.

- **Operating Income Before Depreciation:** Operating Income Before Depreciation (oibdp)/Assets(at). Data is annual and obtained from Compustat. The sample period is from 1993-2013. Book leverage is winsorized at the 5% level.

- **Log Compensation:** Log compensation is the log of total compensation (tdc1). The data is obtained from http://sites.temple.edu/lnaveen/data/. The sample period is from 1993-2010 and the source of the data is Execucomp.

- **Log Delta:** Log Delta is defined as the dollar change in wealth associated with a 1% change in the firms stock price (in$000s). The data is obtained from http://sites.temple.edu/lnaveen/data/. The sample period is from 1993-2010 and the source of the data is Execucomp.

- **Log Vega:** Log Vega is defined as dollar change in wealth associated with a 0.01 change in the standard deviation of the firms returns (in $000s). The data is obtained from http://sites.temple.edu/lnaveen/data/. The sample period is from 1993-2010 and the source of the data is Execucomp.